ECA08 – ANALOG AND DIGITAL COMMUNICATION
CO4 INVESTIGATIVE ASSESSMENT

Modulation and Multiplexing for a Rural Digital Radio Link: An Engineering Investigation

Course Outcome – CO4
Evaluate digital modulation techniques including ASK, FSK, PSK (BPSK, QPSK), QAM for 4G/5G, continuous phase modulation, signal space representation with trade-offs, BER performance, and multiplexing methods such as TDM, FDM, digital multiplexers, and switching systems.

Nature of Assessment: Open-Ended Engineering Investigation Assignment
Duration: 2 Weeks
Maximum Marks: 100
Primary Bloom's Levels: K4 – Analyze | K5 – Evaluate | K6 – Create/Design

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THE ENGINEERING INVESTIGATION PROBLEM
A rural connectivity programme is linking a cluster of villages to the nearest optical-fibre point-of-presence (PoP) using a licensed point-to-point digital microwave link. A village aggregation node collects all local traffic, multiplexes it into a single digital stream and transmits it to the PoP.
The aggregation node must carry the following traffic:

Traffic Source	Quantity	Rate per Source	Service Requirement
Digitised voice (PCM)	30 channels	64 kbps	Real-time; must be carried at all times
School and primary health centre data	4 streams	2 Mbps ± 50 ppm (independent clocks)	Carry as many streams as conditions permit
Agricultural IoT sensor data (aggregated at the node)	120 sensors	4.8 kbps	Delay-tolerant; must be carried at all times

The backhaul link from the node to the PoP has the following characteristics:

Parameter	Value
Allocated RF channel bandwidth	5 MHz
Pulse shaping	Raised-cosine spectrum, roll-off factor $\alpha = 0.25$

Received carrier-to-noise-density ratio C/N_0 – clear sky (at least 99% of the time)	91 dB-Hz
Received C/N_0 – worst-case monsoon rain fade (up to 1% of the time)	82 dB-Hz
Target bit error rate (BER) at demodulator output	$\leq 10^{-6}$
Minimum operating margin required by the operator	3 dB above the E_b/N_0 needed for the target BER
Channel model	Additive white Gaussian noise (AWGN); multipath fading may be neglected
Transmitter power amplifier	Linear, operated with back-off

A second, short-range link connects the 120 sensors to the aggregation node:

Parameter	**Value**
Sensor transmitter	Battery-powered; low-cost saturated (non-linear) power amplifier
Data rate per sensor	4.8 kbps
Frequency band shared by all sensors	1 MHz
Target BER	$\leq 10^{-3}$ (lost packets are re-sent)
Receiver preference	Low cost; non-coherent detection preferred where feasible

The case does not specify the modulation scheme for the backhaul and sensor links, the rain-fade mitigation strategy, the multiplexing method, frame structure, or node switching arrangement. These choices must be investigated by comparing suitable alternatives and supported with analysis and simulation evidence. Each selected approach should be justified based on the observed results.

The assessment focuses on investigating the suitability and limitations of different modulation and multiplexing choices under varying link conditions, determining when the selected approaches may become ineffective, and recommending appropriate actions for reliable system operation.

STAGE 1 – INTERPRET THE PROBLEM AND PLAN THE INVESTIGATION

10 Marks | K4 – Analyze

Analyse the problem before beginning any detailed calculation or simulation.

Determine and discuss:

1. Total payload bit rate that the backhaul must carry when all traffic is present.
2. Payload bit rate of the traffic that must be carried at all times.
3. Maximum symbol rate that the 5 MHz channel can support with the specified pulse shaping.
4. Minimum bandwidth efficiency (bit/s/Hz) needed to carry:
   (a) the full traffic;
   (b) the traffic that must be carried at all times.
5. The relationship between C/N_0, bit rate and E_b/N_0, and why the available E_b/N_0 is different for each modulation scheme even though C/N_0 is fixed.
6. The constraints imposed by the sensor link that do not apply to the backhaul link.

Stage 1 Deliverable

Prepare a concise Investigation Plan containing:

- at least three research questions that this assignment must answer (for example: "Which modulation schemes can meet the BER target in clear sky with the required margin?");
- a hypothesis for each research question, written before you carry out the analysis;
- for each planned experiment, the independent variable, dependent variable and controlled variables;
- the method (analytical, simulation or literature) you will use to answer each question; and
- the evidence that will confirm or reject each hypothesis.

Note: Hypotheses will not be marked on whether they turn out to be correct. They will be marked on whether they are clear, testable and honestly revisited in later stages.

STAGE 2 – INVESTIGATE THE CANDIDATE SCHEMES IN SIGNAL SPACE

15 Marks | K4/K5 – Analyze and Evaluate

Investigate these four representative candidate schemes for the backhaul link:

- BPSK;
- QPSK;
- 8-PSK;
- 8-QAM (state the constellation you choose);
- 16-QAM; and
- 64-QAM.

For each candidate:

1. Identify the basis functions and draw the signal-space (constellation) diagram with Gray-coded bit labels.
2. Determine the minimum Euclidean distance d_{min}, normalised to the same average energy per bit E_b.
3. Determine the peak-to-average power ratio of the constellation (ignoring pulse shaping).
4. Determine the bandwidth efficiency in bit/s/Hz with $\alpha = 0.25$.
5. Determine, from theory, the E_b/N_0 required to achieve BER = 10^{-6} in AWGN. For 8-QAM, a nearest-neighbour approximation or a simulation result may be used.

Present a comparison table of all candidates.

Investigative Question

Using signal-space reasoning, explain why 16-QAM requires less E_b/N_0 than 16-PSK for the same BER, even though both carry four bits per symbol. Support your explanation with calculated d_{min} values.

Stage 2 Deliverable

A ranked shortlist of candidate schemes, with a clear statement of the power-efficiency versus bandwidth-efficiency trade-off revealed by your comparison.

STAGE 3 – INVESTIGATE LINK FEASIBILITY AND SELECT THE MODULATION STRATEGY

15 Marks | K5 – Evaluate

For each of the four candidate schemes, determine for both clear-sky and rain-fade conditions:

- the maximum bit rate the 5 MHz channel can carry;
- whether that bit rate is sufficient for the full traffic and for the traffic that must be carried at all times;
- the E_b/N_0 available at the receiver;
- the operating margin relative to the E_b/N_0 required for BER = 10^{-6}; and
- whether the operator's 3 dB margin requirement is met.

Summarise your results in a feasibility matrix (scheme × condition), marking each entry as feasible, marginal or not feasible.

Investigate whether a single fixed modulation scheme can satisfy all requirements, or whether the link must change its modulation when conditions change (adaptive modulation).

Design Decision 1

Complete the statement:

I recommend _____ for clear-sky operation because _____.

Design Decision 2

Complete the statement:

During rain fade, I recommend _____, carrying _____ traffic, because _____.

You must state which traffic is dropped during rain fade and explain why the alternative strategies were rejected.

Note: Your investigation may show that no scheme meets every requirement during rain fade. If so, say so clearly, quantify the shortfall in dB, and justify the compromise you recommend. A well-supported compromise will receive more credit than an unsupported claim that all requirements are met.

STAGE 4 – DESIGN THE MULTIPLEXING AND SWITCHING ARRANGEMENT

15 Marks | K5/K6 – Evaluate and Design

Investigate the suitability of:

- Time Division Multiplexing (TDM), in which all traffic is interleaved into one digital stream on a single carrier; and
- Frequency Division Multiplexing (FDM), in which the 5 MHz channel is divided into separate sub-channels for different traffic types.

Compare the two methods considering:

- guard-band and framing overhead;
- overall spectral efficiency;
- flexibility when the link changes modulation during rain fade;
- synchronisation requirements; and
- implementation complexity.

Design Decision 3

Complete the statement:

I recommend _____ multiplexing because _____.

Using your selected method, develop the multiplexing arrangement:

1. Design a frame structure. Specify the frame duration, the number of time slots (or sub-channels), the allocation of slots to each traffic type, the frame-alignment and signalling overhead, and the resulting line bit rate.
2. Show that the line bit rate fits within the bit rate of the modulation selected in Design Decision 1.
3. Show how the frame is re-mapped during rain fade, identifying which slots are retained and which are dropped.
4. The four data streams arrive with independent clocks of 2 Mbps ± 50 ppm. Explain how your digital multiplexer accommodates these clock differences, and calculate the justification (bit-stuffing) capacity required per stream.

Switching at the Node

To allow re-routing during faults, the node must be able to connect any of the 30 incoming voice channels to any voice time slot of the outgoing frame.

1. Determine the memory size and the maximum permissible memory access time for a time-slot interchange (TSI) switch that performs this function.
2. Estimate the number of crosspoints that an equivalent space-division switch would need, and compare the two switching approaches.

Draw a clearly labelled frame diagram and a block diagram of the complete multiplexing and switching arrangement.

STAGE 5 – SIMULATION-BASED BER INVESTIGATION

20 Marks | K4/K5 – Analyze and Evaluate

Use MATLAB, Python or LabVIEW to carry out the following experiments.

Experiment A – Validation in AWGN

1. Develop a Monte Carlo simulation (bit source → modulator → AWGN channel → demodulator → error counter) for at least three shortlisted schemes, including your clear-sky and rain-fade choices.
2. Plot simulated BER against E_b/N_0 (0–20 dB) and overlay the theoretical curves on the same axes.
3. Quantify any difference between simulation and theory and explain its cause.
4. Justify the number of bits simulated at each E_b/N_0 point. Explain how many bit errors are needed for a reliable BER estimate, and how you verified performance at 10^{-6}, where direct simulation becomes expensive.

Experiment B – Effect of a Practical Impairment

Practical receivers and transmitters are never ideal. Select **one** of the following impairments:

- (a) Carrier phase offset – a constant phase error θ caused by imperfect carrier recovery. Investigate three representative phase offsets: θ = 0°, 10° and 20°.
- (b) Amplitude compression – a power amplifier driven close to saturation, modelled as a soft limiter that clips the symbol amplitude at a chosen fraction of the peak constellation amplitude. Investigate three clipping levels.

Before simulating, state your hypothesis: which of your shortlisted schemes will be most affected, and why, based on its signal-space geometry. State clearly how you define E_b/N_0 in the presence of the impairment.

Then:

1. Simulate BER against Eb/N0 for each selected impairment level.
2. Plot received constellation (scatter) diagrams at E_b/N_0 = 15 dB, with and without the impairment.
3. Determine the E_b/N_0 penalty (in dB) at BER = 10^{-4} for each impairment level.

4. Identify the largest tested impairment level that keeps the penalty below 1 dB for each scheme, where applicable.
5. State whether your hypothesis was confirmed or rejected, and explain why.

Finally, re-examine Design Decisions 1 and 2. Does the impairment experiment change your margin calculation or your recommendation?

Permitted Use of Computational Tools

The Monte Carlo simulation (bit generation, symbol mapping, noise addition, detection and error counting) must be your own implementation. To keep the task achievable within one week, simulate three schemes: your proposed clear-sky scheme, your proposed rain-fade scheme, and one alternative for comparison.

Built-in functions such as MATLAB berawgn or equivalent library routines may be used only to generate theoretical reference curves.

Simulation plots without an explanation of what they show and what they imply for the design will not receive full marks.

STAGE 6 – INVESTIGATE THE SENSOR LINK: CONSTANT-ENVELOPE MODULATION

10 Marks | K5 – Evaluate

The sensor transmitters use low-cost saturated power amplifiers and run on batteries.

Investigate the suitability of:

- Amplitude Shift Keying (ASK / on-off keying);
- non-coherent Binary Frequency Shift Keying (BFSK);
- Minimum Shift Keying (MSK), as an example of continuous phase modulation; and
- BPSK.

Compare the candidates considering:

- whether the signal envelope is constant, and what that means for a saturated amplifier;
- coherent or non-coherent detection, and its effect on receiver cost;
- E_b/N_0 required for BER = 10^{-3};
- main-lobe bandwidth and side-lobe decay of the power spectral density; and
- energy consumed per transmitted bit (battery impact).

Generate or obtain power spectral density plots for BFSK and MSK at the same bit rate. Use the plots to explain the benefit of phase continuity.

Investigate whether the 120 sensors should share the 1 MHz band by TDM (time slots) or FDM (separate frequency channels), considering the bandwidth occupied by your chosen modulation and the timing or frequency accuracy that each method demands from low-cost sensors.

Design Decision 4

Complete the statement:

I recommend _____ with _____ multiplexing for the sensor link because _____.

STAGE 7 – INVESTIGATE QAM IN 4G AND 5G SYSTEMS

5 Marks | K4/K5 – Analyze and Evaluate

Your adaptive backhaul design uses the same principle that 4G LTE and 5G NR use to cope with changing channel conditions. Using at least two credible sources, briefly investigate:

1. Which modulation orders 4G LTE and 5G NR support in the downlink and the uplink.

2. How a 4G/5G base station decides which modulation to use, through channel quality reporting and the selection of a modulation and coding scheme (MCS).
3. Why 5G NR includes π/2-BPSK for the uplink, and how this connects to your findings in Stage 6.
4. One important difference between 4G/5G link adaptation and the adaptive strategy you proposed in Stage 3.

Cite at least two credible sources (for example: 3GPP technical specifications, IEEE papers, or standard textbooks). Write in your own words; copied text will not receive credit.

FINAL ENGINEERING DECISION

10 Marks | K5/K6 – Evaluate and Create

WOULD YOU RECOMMEND COMMISSIONING THIS LINK DESIGN?

State clearly:

YES / NO

Defend your decision using evidence gathered across the complete investigation. Your conclusion must consider:

- BER performance and operating margin in clear sky and during rain fade;
- which services survive rain fade and which do not;
- spare capacity for future traffic growth;
- sensitivity to receiver and transmitter impairments (Stage 5);
- complexity of adaptive modulation and of the multiplexing and switching arrangement; and
- battery life and cost of the sensor nodes.

Present a final table:

Requirement	Specified	Achieved (Clear Sky)	Achieved (Rain Fade)	Satisfied?
Payload bit rate carried	Full traffic			
Voice and IoT traffic carried	At all times			
Number of data streams carried	4			
Backhaul BER	$\leq 10^{-6}$			
Operating margin	≥ 3 dB			
Sensor link BER	$\leq 10^{-3}$			

If your answer is NO

Propose a **redesign**.

Identify exactly what you would change, such as:

- the set of modulation schemes;
- the allocated channel bandwidth or the roll-off factor;
- antenna gain or transmit power (to raise C/N_0);

- the addition of forward error correction;
- traffic priorities; or
- the operator's margin requirement.

Explain why you would make the change and quantify its expected effect – for example, the additional C/N_0 (in dB) needed for your preferred scheme to meet the 3 dB margin during rain fade.

One-week scope guidance: Use concise calculations and comparison tables. For simulations, focus on the three shortlisted backhaul schemes and one selected impairment; include only plots that directly support a design decision. Keep the final report focused on evidence, trade-offs and recommendations.

SUBMISSION REQUIREMENTS

Submit one concise Engineering Investigation Report, prioritising essential calculations, comparison tables, clearly labelled plots and justified engineering decisions over lengthy background descriptions.

Your report should demonstrate the complete investigation process:

PROBLEM → RESEARCH QUESTIONS → HYPOTHESES → ANALYSIS → SIMULATION → INTERPRETATION → DECISION → RECOMMENDATION

The submission must contain:

- the concise Investigation Plan (Stage 1);
- signal-space diagrams and the candidate comparison table;
- the feasibility matrix;
- the frame structure, multiplexer and switching diagrams;
- BER plots with theoretical overlays, and constellation diagrams;
- power spectral density plots for the sensor link;
- a reference list for Stage 7;
- relevant code as an appendix; and
- the final engineering recommendation.

ASSESSMENT RUBRIC

Assessment Criterion	**Marks**	**Primary BL**
Interpretation of the problem and investigation plan	10	K4
Signal-space investigation of candidate schemes	15	K4/K5
Link feasibility and selection of modulation strategy	15	K5
Multiplexing and switching design	15	K5/K6
Simulation-based BER investigation	20	K4/K5

Constant-envelope investigation for the sensor link	10	K5
Investigation of QAM in 4G/5G systems	5	K4/K5
Final engineering decision and report	10	K5/K6
TOTAL	**100**	

Results without corresponding engineering reasoning will not receive full marks.

PERFORMANCE LEVEL RUBRICS

Criteria	Maximum Marks	Excellent	Good	Satisfactory	Needs Improvement
1. Problem Interpretation and Investigation Plan	10	Correctly calculates full and always-on payload rates, maximum symbol rate and required spectral efficiencies; explains C/N_0-to-Eb/N_0 relationships and distinct sensor-link constraints. Provides at least three clear, testable research questions and hypotheses, identifies variables, methods and evidence, and revisits hypotheses honestly. (9–10)	Calculations and investigation plan are mostly sound; questions, variables or evidence criteria have minor gaps. (7–8)	Identifies the main requirements but misses some calculations or gives vague hypotheses, variables or methods. (5–6)	Misinterprets major requirements; plan is missing, untestable or lacks meaningful calculations. (0–4)
2. Signal-Space and Candidate-Modulation Analysis	15	Accurately presents labelled Gray-coded constellations and basis functions for	Most diagrams and calculations are correct; comparison and ranking are well	Provides basic constellation diagrams or calculations, but several metrics or the technical	Constellations are missing/incorrect; major errors in distance, PAPR, efficiency or

		the selected candidates; calculates/compares normalized minimum Euclidean distance, PAPR, spectral efficiency and required E_b/N_0 for BER 10^{-6}. Clearly explains the power-versus-bandwidth trade-off and supports ranking with evidence. (14–15)	supported with minor omissions. (11–13)	justification are incomplete. (8–10)	BER reasoning. (0–7)
3. Backhaul Feasibility and Adaptive-Modulation Strategy	15	Evaluates each shortlisted scheme under clear-sky and rain-fade C/N_0; calculates achievable bit rate, available E_b/N_0 and margin against BER and the required 3 dB margin. Completes a clear feasibility matrix, identifies retained/dropped traffic during fade, quantifies any shortfall and justifies adaptation or compromise. (14–15)	Feasibility calculations and modulation recommendations are mostly correct, with minor gaps in margin analysis or traffic prioritisation. (11–13)	Provides partial calculations or a basic recommendation, but the feasibility matrix, rain-fade response or shortfall analysis is incomplete. (8–10)	Claims feasibility without evidence or makes major errors in C/N_0, E_b/N_0, BER, capacity or traffic decisions. (0–7)
4. Multiplexing, Frame Structure and Node Switching Design	15	Justifies TDM or FDM using overhead, spectral efficiency, synchronization, flexibility and complexity. Defines frame duration,	Architecture, framing and switching calculations are mostly sound, with minor omissions in overhead, clock	Provides a basic multiplexing design but frame details, bit-stuffing, switching calculations or fade remapping are incomplete. (8–10)	Frame/switching design is missing or technically inconsistent; calculations and architecture are unsupported. (0–7)

		slots/subchannels, traffic allocation, signalling and line rate; addresses ±50 ppm clock differences and bit-stuffing capacity; correctly analyses TSI memory/access time and space-switch crosspoints. Includes clear labelled diagrams and rain-fade remapping. (14–15)	accommodation or comparison. (11–13)		
5. BER Simulation and Practical-Impairment Investigation	20	Implements an original Monte Carlo chain for at least three shortlisted schemes; plots simulated and theoretical BER over E_b/N_0, justifies trial counts and treatment of BER 10^{-6}, and quantifies discrepancies. Investigates the selected phase-offset or amplitude-compression impairment with labelled BER and constellation plots, computes E_b/N_0 penalties and tests the hypothesis. Revises recommendations using results. (18–20)	Simulation is valid and includes most required plots and comparisons; minor gaps in statistical justification, penalty estimates or interpretation. (14–17)	Simulation results are partial; theoretical comparison, trial-count justification, impairment analysis or design implications are weak. (10–13)	Little valid simulation evidence; plots are missing/unlabelled, code is not meaningfully implemented, or results are not interpreted. (0–9)

6. Sensor-Link Modulation and Multiplexing Selection	10	Compares ASK/OOK, non-coherent BFSK, MSK and BPSK for envelope, saturated-amplifier compatibility, detection cost, E_b/N_0 at BER 10^{-3}, PSD, spectral occupancy and battery energy. Uses BFSK/MSK PSD plots to explain phase continuity and selects a justified TDM/FDM approach for 120 sensors within 1 MHz. (9–10)	Comparison and recommendation are mostly correct, with minor omissions in PSD interpretation, energy use or multiplexing constraints. (7–8)	Provides basic comparisons but misses key quantitative evidence or weakly justifies sensor modulation/multiplexing. (5–6)	Major errors in modulation properties or no feasible, justified sensor-link design. (0–4)
7. Investigation of QAM in 4G/5G Systems	5	Uses at least two credible sources to accurately describe supported modulation orders, CQI/MCS-based link adaptation, the purpose of $\pi/2$-BPSK uplink use, and one meaningful difference from the proposed backhaul adaptation. References are cited and explained in the student's own words. (5)	Sources are credible and most technical points are correct, with a minor omission or citation weakness. (4)	Briefly addresses the topic but lacks detail, has limited source quality or incompletely connects findings to the design. (3)	Claims are inaccurate, sources are absent/unreliable, or copied material is not properly acknowledged. (0–2)

8. Final Engineering Decision and Specification Verification	10	States a clear YES/NO commissioning recommendation and completes a consistent specification-verification table for clear sky and rain fade. Integrates BER/margin, service continuity, spare capacity, impairment sensitivity, implementation complexity, battery life and cost. If requirements fail, proposes a specific redesign and quantifies its expected effect. (9–10)	Decision is supported by most evidence and the verification table is largely complete; minor gaps in redesign or trade-off discussion. (7–8)	Provides a basic decision, but the verification table, cross-stage evidence or quantified redesign is incomplete. (5–6)	Decision is unsupported or missing; important requirements are not checked and no credible redesign is given. (0–4)